

Jamie Hayes

CONTACT INFORMATION

Email: jamiedhayes1@gmail.com

Website: My research scholar page

EXPERIENCE

Google DeepMind, London, UK

Staff Research Scientist

May, 2025 - Present

- Research lead for Gemini Security Modelling, a team of 9 members of technical staff working on evaluating and improving Gemini's resilience to prompt injection and jailbreak attacks in security-sensitive settings.

Google DeepMind, London, UK

Senior Research Scientist

May, 2023 - May, 2025

- Proposed and led watermarking project, which then became SynthID: <https://deepmind.google/technologies/synthid/>. This is Google's provenance detection system that is deployed across all generative models available to Google customers.
- AI agents that have the ability to act upon untrusted data (for example, a model that can read and write emails) are vulnerable to "indirect prompt injection", an attack where the model may share private user data with an untrusted party. I lead a team developing internal benchmarks and auditing techniques to measure the vulnerability to indirect prompt injection attacks, and more generally our remit is to design Trustworthy & Capable AI Agents. Our work will be deployed across all models developed by Google, reducing the risk of leaking private user data.

Google DeepMind, London, UK

Research Scientist

June, 2020 - May, 2023

Research into verification and robustness guarantees in machine learning. Developed techniques to audit privacy-preserving ML pipelines – now deployed in Google Photos.

Google DeepMind, London, UK

Internship

May, 2019 - September, 2019

Research into watermarking applications in machine learning. This eventually led to the formation of the SynthID project.

Google Research, Mountain View, CA, USA

Internship

July, 2018 - September, 2018

Research into generative and adversarial machine learning. Developed new techniques for unsupervised style transfer, and new probabilistic conditioning methods for generative models.

Microsoft Research, Cambridge, UK

Internship

February, 2018 - May, 2018

Research into security and privacy attacks and defenses in multi-party machine learning.

Naval Research Laboratory, Washington, DC, USA

Internship

August, 2017 - October, 2017

Research into adversarial machine learning and network traffic analysis. Developed a project using machine learning to perform privacy-preserving experiments on live traffic analysis on Tor.

Government Digital Services, London, UK

Intern for the SecOps team

March, 2017 - July, 2017

I developed and implemented a privacy-preserving machine learning pipeline to aid threat analysis and improve Transaction Monitoring (TxM) on the GOV.UK Verify system.

University of Manchester, Manchester, UK

Network Technician

September, 2011 - March, 2013

Maintenance of the internal University network that provided a connection to over 10,000 students. Duties included server maintenance, switch configurations, some SDN programming.

EDUCATION

Dept. of Computer Science, University College London, UK

Sep. 2014 - June. 2020 - Ph.D. candidate

Privacy, Security and Machine Learning

Advisor: Prof. George Danezis

Second Advisor: Prof. Thore Graepel

Computer Laboratory, University of Cambridge, UK

Sep. 2013 - Apr. 2014 - Researcher

Parameterized Computational Complexity of the Graph Isomorphism Problem

Advisor: Prof. Anuj Dawar

Dept. of Mathematics, University of Manchester, UK

Sep. 2007 - Jun. 2011 - Master of Mathematics

First Class Honours

Advisor: Prof. Richard Sharp

SKILLS

Knowledge of:

Python • Go • Shell • $\text{\LaTeX}$ • Unix/Linux • Vim • Machine Learning (TensorFlow, PyTorch, JAX, SKLearn) • C++ • Git

Exposure to:

Haskell • JavaScript • C • SQL • HTML • CSS

CODE AND SIDE PROJECTS

Public code available at <https://github.com/jhayes14>, code for private projects available on request.

REFERENCES

Available upon request.

SERVICES

- Reviewing for most major academic ML and Security conferences since 2015 (IEEE S&P, CCS, USENIX Security, PETS, ICML, ICLR, NeurIPS etc.). Full breakdown available upon request.
- Organizer of the workshop Synthetic Data Generation: Quality, Privacy, Bias at ICLR 2021.
- Various invited talks and lectures at conferences, academic labs, summer schools etc. Full breakdown available upon request.